



Founded 1982

STUDENT NUMBER: _____

TEACHER: _____

THE HILLS GRAMMAR SCHOOL

Trial Higher School Certificate Examination 2015

MATHEMATICS EXTENSION 1

Time Allowed: Two hours (plus five minutes reading time)

Weighting: 40%

Outcomes: H6, H7, H8, H9, HE1, HE2, HE4, HE7, HE9

General Instructions:

- Board-approved calculators may be used
- Attempt all questions
- Start all questions on a new sheet of paper
- The marks for each question are indicated on the examination
- Show all necessary working for Questions 11-14
- The diagrams are not drawn to scale
- A table of standard integrals is provided

Total Marks – 70

Section I Questions 1-10

10 Marks

Allow about 15 minutes for this section

Section II Questions 11-14

60 Marks

Allow about 1 hour and 45 minutes for this section

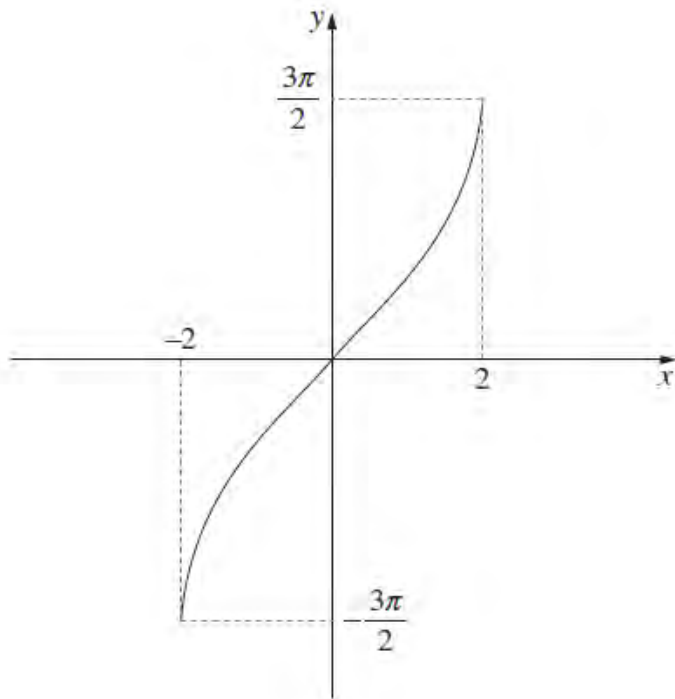
MCQ	Question 11	Question 12	Question 13	Question 14	TOTAL
10	15	15	15	15	70

Section 1 Multiple Choice (10 Marks)

1 Given that $\tan\left(\frac{\theta}{2}\right) = t$, then $\sin \theta$ would be written as:

- (A) $\frac{2t}{1-t^2}$
 (B) $\frac{1-t^2}{1+t^2}$
 (C) $\frac{1+t^2}{1-t^2}$
 (D) $\frac{2t}{1+t^2}$

2 Which function best describes the following graph:



- (A) $y = 3\sin^{-1} 2x$
 (B) $y = \frac{3}{2}\sin^{-1} 2x$
 (C) $y = 3\sin^{-1} \frac{x}{2}$
 (D) $y = \frac{3}{2}\sin^{-1} \frac{x}{2}$

3 Evaluate $\sum_{n=3}^{10} 8 + 5n$

(A) 283.5

(B) 324

(C) 567

(D) 648

4 The interval AB is divided internally in the ratio 3:1 by the point $P(x, y)$. Given $A(-7, 7)$ and $B(1, -5)$ then the values of x and y are:

(A) $x = -2$ and $y = 2$

(B) $x = 2.5$ and $y = 2$

(C) $x = -1$ and $y = -2$

(D) $x = 1$ and $y = -2$

5 Which expression is the correct factorisation of $x^3 - 27$?

(A) $(x - 3)(x^2 - 3x + 9)$

(B) $(x - 3)(x^2 - 6x + 9)$

(C) $(x - 3)(x^2 + 3x + 9)$

(D) $(x - 3)(x^2 + 6x + 9)$

6 The parametric equation of a function is:

$$x = 2t^2, y = 4 - t.$$

The Cartesian equation is

(A) $x = 4(2 - y)^2$

(B) $x = 2(y - 4)^2$

(C) $x = 2(y + 4)^2$

(D) $x = 2(4 - y)^2$

7 Evaluate $\lim_{x \rightarrow 0} \frac{x}{\sin 2x}$:

(A) 0

(B) 0.5

(C) ∞

(D) 2

8 Which expression is equal to $\int \sin^2 3x \, dx$:

(A) $\frac{1}{2} \left(x - \frac{1}{3} \sin 3x \right) + C$

(B) $\frac{1}{2} \left(x + \frac{1}{3} \sin 3x \right) + C$

(C) $\frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right) + C$

(D) $\frac{1}{2} \left(x + \frac{1}{6} \sin 6x \right) + C$

9 A particle is moving in simple harmonic motion with displacement x . Its velocity v is given by

$$v^2 = 16(9 - x^2)$$

What is the amplitude, A , and the period, T , of the motion?

(A) $A = 3$ and $T = \frac{\pi}{2}$

(B) $A = 3$ and $T = \frac{\pi}{4}$

(C) $A = 4$ and $T = \frac{\pi}{3}$

(D) $A = 4$ and $T = \frac{2\pi}{3}$

10 The polynomial $P(x) = x^3 + ax^2 + ax + 1$ leaves a remainder of 3 when divided by $(x - 2)$.

The value of a is:

(A) 1

(B) -1

(C) -3

(D) 3

END OF SECTION 1

Section 2**Marks****BEGIN A NEW BOOKLET****Question 11 (15 marks)**

(a) Find $\frac{d^2}{dx^2} e^{x^2}$. **2**

(b) Find k such that $\int_1^k (3 - \frac{1}{x^2}) dx = 0$. **3**

(c) Use the substitution $u = 1 + e^x$ to evaluate $\int_0^{\ln 2} \frac{e^x}{(e^x + 1)^2} dx$. **4**

(d) Let $I = \int_0^{\frac{\pi}{2}} \cos^2 x \, dx$.

(i) Find, by integration, the exact value of I . **2**

(ii) Use Simpson's rule with 3 function values to approximate I . **2**

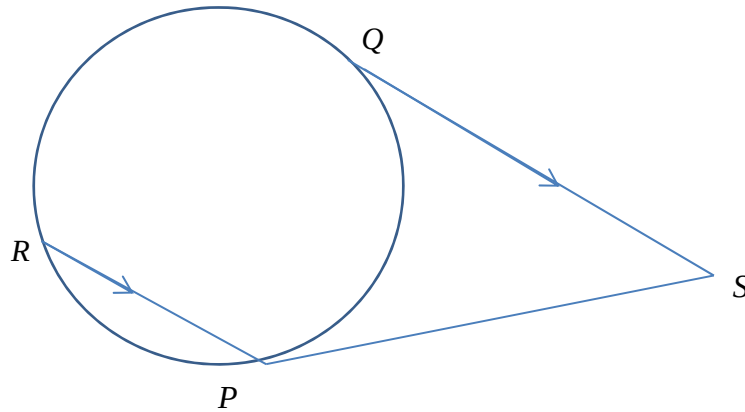
(e) i) Show that $e^{x \ln 2} = 2^x$. **1**

ii) Hence find $\frac{d}{dx} 2^x$ **1**

Question 12 (15 marks) BEGIN A NEW BOOKLET

- (a) In the diagram the points P and Q lie on a circle and the tangents to the circle at P and Q meet at S .

R is a point on the circle so that RP is parallel to QS .



Copy or trace the diagram into your writing book.

- i) Explain why $\triangle PSQ$ is isosceles, 2
 - ii) Show that $\triangle PQR$ is isosceles, 2
 - iii) Deduce that $QP = QR$. 1
- (b) Detective Angela Baker is called to a murder scene at 3:27a.m. She measures the victim's body temperature at that time to be 27°C and one hour later it has dropped to 25°C . The cooling rate of the body is proportional to the difference between the room temperature 21°C and the temperature T , of the body. That is, T satisfies the equation

$$\frac{dT}{dt} = -k(T - 21) \quad \text{where } k \text{ is a positive constant, and } t \text{ is the number of hours after 3:27a.m.}$$

- (i) Verify that $T = 21 + Ae^{-kt}$ is a solution of this equation, where A is a constant. 1
- (ii) Find the exact values of A and k . 2
- (iii) Assuming that the victim's body temperature was 37°C at the time of death, when was the murder committed?
Give your answer to the nearest minute. 2

Question 12 continued

- (c) If α , β , and γ are the roots of the equation $2x^3 - x^2 - 5x + 6 = 0$
find the value of $\alpha^2 + \beta^2 + \gamma^2$. 2

- (d) Use mathematical induction to show that for all integers $n \geq 1$,

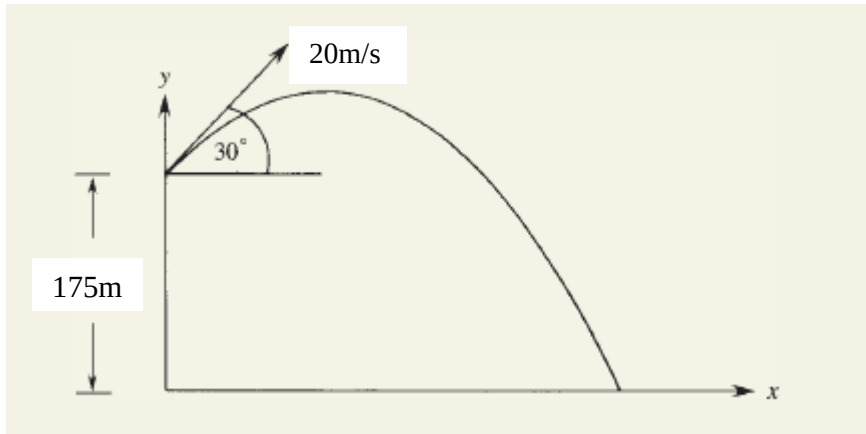
$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}.$$
 3

Question 13 (15 marks) BEGIN A NEW BOOKLET

- (a) (i) Prove, using calculus, that the equation $x^3 + 2x + 4 = 0$ has only one real root α . 2
- (ii) Show that $-2 < \alpha < -1$. 1
- (iii) Starting with an initial approximation of $\alpha = -1$, use one application of Newton's method to find a further approximation for α . 2
- (b) A particle is moving in simple harmonic motion along the x -axis. Its velocity v , at x , is given by $v^2 = 24 - 8x - 2x^2$.
- (i) Find all values of x for which the particle is at rest. 2
- (ii) Find an expression for the acceleration of the particle, in terms of x . 1
- (iii) Find the maximum speed of the particle. 2

Question 13 continued

- (c) A man who is standing on top of a vertical cliff throws a stone into the air at an angle θ to the horizontal. The top of the cliff is 175 metres above a flat sea.



The initial velocity of the stone is 20ms^{-1} . Acceleration due to gravity is -10ms^{-2} .
The path of the stone is given by the parametric equations

$$x = 20t \cos \theta \quad \text{and} \quad y = 20t \sin \theta - 5t^2 + 175$$

The angle of projection of the stone to the horizontal is 30° .

- | | |
|--|----------|
| (i) Find the time it takes for the stone to hit the water. | 3 |
| (ii) Find the speed at which the stone hits the water. | 2 |

Question 14 (15 marks) BEGIN A NEW BOOKLET

- | | |
|--|----------|
| (a) (i) Write $\cos x - \sqrt{3} \sin x$ in the form $R \cos(x + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$, | 2 |
|--|----------|

- | | |
|--|----------|
| (ii) Hence, or otherwise, solve the equation $\cos x - \sqrt{3} \sin x = 1$ for $0 \leq x \leq 2\pi$. | 2 |
|--|----------|

Question 14 continued

(b) The point $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$.

(i) Show that the equation of the tangent to $x^2 = 4ay$ at P is $px - y - ap^2 = 0$. 2

(ii) The tangent at P cuts the x axis at X . Find the coordinates of X . 1

(iii) Show that PX is perpendicular to SX , where S is the focus of the parabola. 2

(iv) A circle is drawn through the points S , X , and P . Show that the coordinates of the centre of the circle are given by

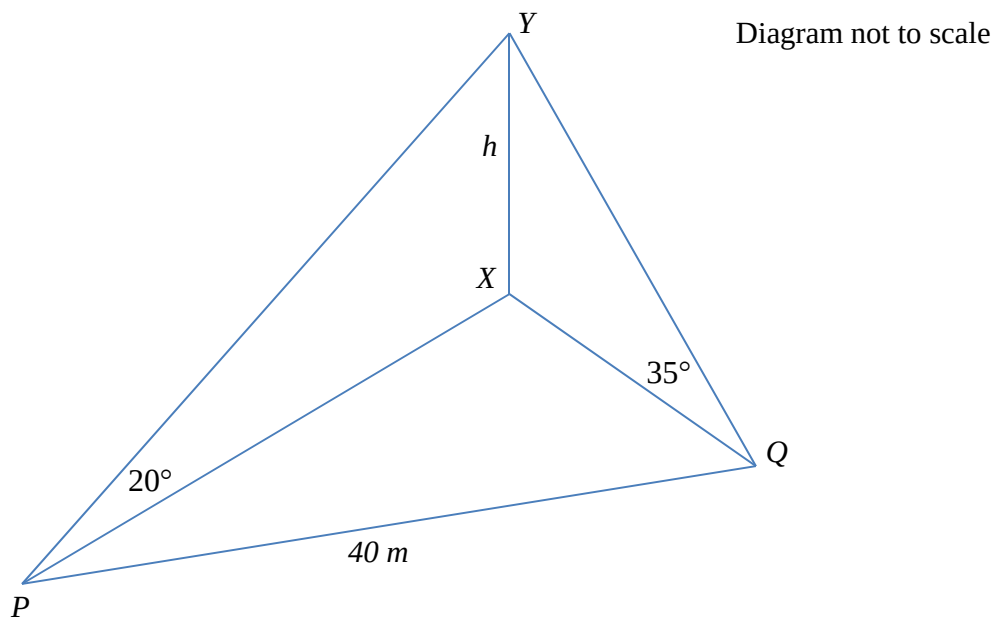
$$C = \left(ap, \frac{a(1+p^2)}{2} \right).$$

Justify your answer.

2

Question 14 continued

- (c) From a point P due south of a vertical tower, the angle of elevation of the top of the tower is 20° and from a point Q due east of the tower it is 35° .
The distance from P to Q is 40 metres.



- | | |
|---|---|
| (i) Find an expression for PX in terms of h . | 1 |
| (ii) Find an expression for QX in terms of h . | 1 |
| (iii) Calculate the height of the tower to the nearest metre. | 2 |

END OF ASSESSMENT TASK

ANSWER SHEET FOR MULTIPLE CHOICE SECTION

Student Exam number: _____

Teacher: _____

1. A ○ B ○ C ○ D ○

2. A ○ B ○ C ○ D ○

3. A ○ B ○ C ○ D ○

4. A ○ B ○ C ○ D ○

5. A ○ B ○ C ○ D ○

6. A ○ B ○ C ○ D ○

7. A ○ B ○ C ○ D ○

8. A ○ B ○ C ○ D ○

9. A ○ B ○ C ○ D ○

10. A ○ B ○ C ○ D ○

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

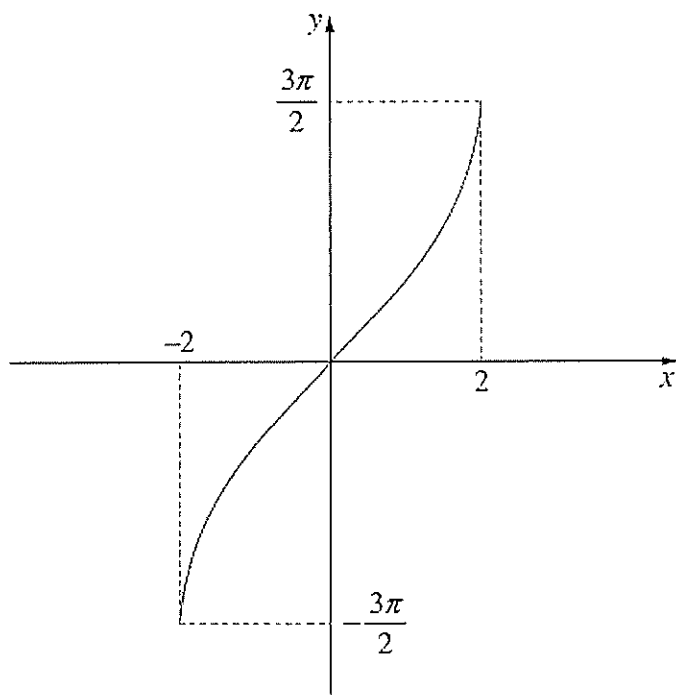
NOTE : $\ln x = \log_e x, \quad x > 0$

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(A) 0

(B) 0.5

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(D) 2

8 Which expression is equal to $\int \sin^2 3x \, dx$:

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☒ (C) $\frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right) + C$

(D) $\frac{1}{2} \left(x + \frac{1}{6} \sin 6x \right) + C$

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The value of a is:

(A) 1

☒ (B) -1

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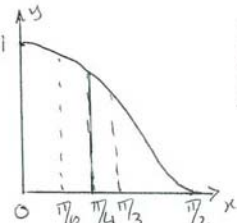
(D) 3

END OF SECTION 1

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) MULTIPLE CHOICE.	comments
1. (D) 2. (C) since for $y = \sin^{-1} x$ $-1 \leq x \leq 1$, $-\pi/2 \leq y \leq \pi/2$ 3. (B) 4. (C) 5. (C) 6. (D) 7. (B) 8. (C) 9. (A) 10. (B)	
QUESTION 11 a) $\frac{d}{dx}(e^{x^2}) = e^{x^2} \cdot 2x$ $\frac{d^2}{dx^2}(e^{x^2}) = \frac{d}{dx}(e^{x^2} \cdot 2x)$ $= e^{x^2} \cdot 2 + 2x \cdot e^{x^2} \cdot 2x$ $= 2e^{x^2}(1 + 2x^2)$ (2) b) $\int_0^k (3 - \frac{1}{x^2}) dx = 0$ $\therefore [3x + x^{-1}]_0^k = 0$ $3k + \frac{1}{k} - (3 + 1) = 0$ $3k^2 + 1 - 4k = 0$ $3k^2 - 4k + 1 = 0$ $(3k - 1)(k - 1) = 0$ $\therefore k = 1 \text{ or } k = \frac{1}{3}$ (3)	good attempt here. poorly done by too many students. failure to recognise a product. — difficulty in dealing with k on the denominator

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) QUESTION 11.	comments								
c) let $u = 1 + e^x$ when $x = 0$, $u = 2$ $du = e^x dx$ $x = \ln 2$, $u = 3$ $\therefore \int_0^{\ln 2} \frac{e^x}{(e^x + 1)^2} dx = \int_2^3 \frac{du}{u^2}$ $= \left[-\frac{1}{u}\right]_2^3$ (4) $= -\frac{1}{3} + \frac{1}{2} = \frac{1}{6}$	Some issues with setup of parameters and du and dx .								
d) i) $I = \int_0^{\pi/2} \cos^2 x dx$ where $\cos^2 x = \frac{1}{2}(\cos 2x + 1)$ $= \frac{1}{2} \int_0^{\pi/2} \cos 2x + 1 dx$ $= \frac{1}{2} \left[\frac{1}{2} \sin 2x + x \right]_0^{\pi/2}$ (2) $= \frac{1}{2} \left[\frac{1}{2} \times 0 + \frac{\pi}{2} \right] = \frac{\pi}{4}$ exact form!	Failure to recognise the need for double angle formula.								
ii)  <table border="1" data-bbox="1520 916 1800 1003"> <tr> <td>x</td><td>0</td><td>$\pi/4$</td><td>$\pi/2$</td></tr> <tr> <td>$\cos^2 x$</td><td>1</td><td>$1/2$</td><td>0</td></tr> </table> $A \doteq \frac{h}{3} [d_F + 4d_m + d_L]$ OR $\doteq \frac{b-a}{6} [f(a) + 4f(\frac{a+b}{2}) + f(b)]$ $= \frac{\pi/2}{12} [1 + 4(\frac{1}{2}) + 0]$ (2) $= 3\pi/12 \text{ or } \pi/4$	x	0	$\pi/4$	$\pi/2$	$\cos^2 x$	1	$1/2$	0	Table of this form is very helpful.
x	0	$\pi/4$	$\pi/2$						
$\cos^2 x$	1	$1/2$	0						
	Some students need to learn the formula.								

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	QUESTION 11.	comments
e) i)	$\begin{aligned} \text{LHS} &= e^{x \ln 2} \\ &= e^{\ln(2^x)} \\ &= 2^x \\ &= \text{RHS} \end{aligned}$	Need to learn index rules for e^x and $\ln x$.
ii)	$\begin{aligned} \frac{d}{dx}(2^x) &= \frac{d}{dx}(e^{x \ln 2}) \\ &= e^{x \ln 2} \cdot \ln 2 \\ &= 2^x \cdot \ln 2 \end{aligned}$	Follows from (i)
QUESTION 12.		
a)		
i)	Join PQ $PS = QS$ (tangents from an external point are equal.) $\therefore \Delta PSQ$ is isosceles (two sides equal)	Need to learn def'n of tangents from an external pt. Identify what makes a triangle isosceles.
ii)	Join QR As ΔPSQ is isosceles, $\angle PQS = \angle QPS$ (base angles of isosceles Δ are equal) $\angle PQS = \angle RPQ$ ($\angle$ between chord and tangent equals $\angle$ in alternate segment) $\angle SQP = \angle RPQ$ (alternate angles from $\parallel$ lines)	ditto definition needed

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	QUESTION 12.	comments
	$\therefore \Delta PQR$ is isosceles (two equal angles)	
iii)	$QP = QR$ (ΔPQR is isosceles; $\angle P, \angle R$ are opposite equal angles)	Explanation.
b)	$\frac{dT}{dt} = -k(T-21)$ i) $T = 21 + Ae^{-kt}$ $\frac{dT}{dt} = 0 + A \cdot e^{-kt} \cdot -k$ $= -k \cdot Ae^{-kt}$ where $T-21 = Ae^{-kt}$ ii) $\frac{dT}{dt} = -k(T-21)$	Verification requiring the derivative and substitution.
	ii) $t=0, T=27$ $27 = 21 + A \rightarrow A = 6$ $t=1, T=25$ $25 = 21 + 6e^{-k}$ $\therefore 6e^{-k} = 4$ $e^{-k} = \frac{2}{3}$ $\therefore -k = \ln\left(\frac{2}{3}\right)$ $k = \ln\left(\frac{3}{2}\right)$	Well done
	iii) When $T=37$, $37 = 21 + 6e^{-\ln(\frac{3}{2})t}$ $16 = 6e^{-\ln(\frac{3}{2})t}$ $\ln\left(\frac{16}{6}\right) = -\ln\left(\frac{3}{2}\right)t$ $t = \ln\left(\frac{16}{6}\right) \div -\ln\left(\frac{3}{2}\right) = -2 \ln 25 \text{ min}$ $\therefore \text{Time} = 3:27 - 2:25 = 1:02 \text{ am}$	Not as well done. Do not write "calculator speak". Express values with hours and minutes.

Reasonably well done.

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
c) $2x^3 - x^2 - 5x + 6 = 0$. $\alpha + \beta + \gamma = \frac{-(-1)}{2}$ $= \frac{1}{2}$ $\alpha\beta + \beta\gamma + \alpha\gamma = \frac{-5}{2}$ $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$ $= \left(\frac{1}{2}\right)^2 - 2\left(-\frac{5}{2}\right)$ $= \frac{1}{4} + 5$ $= 5\frac{1}{4}$ d) Prove: $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$ Test $n=1$. LHS = $\frac{1}{1(2)}$ RHS = $\frac{1}{2}$ LHS = RHS. $\therefore$ true for $n=1$. Assume true for $n=k$. $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1} \quad \text{--- (A)}$ Prove for $n=k+1$. i) prove $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$ LHS = $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)}$ $= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$ $= \frac{k(k+2) + 1}{(k+1)(k+2)} = \frac{(k^2 + 2k + 1)}{(k+1)(k+2)}$ $= \frac{(k+1)^2}{(k+1)(k+2)}$ $= \frac{k+1}{k+2} = \text{RHS}.$	— These need to be learned. — ditto — careful subst'n needed. Some rocky presentations here.

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
If true for $n=k$, then true for $n=k+1$. $\therefore$ Statement is true for $n=1, 2, 3, \dots$ and all positive integral n. QUESTION 13. a) i) $y = x^3 + 2x + 4$ $\frac{dy}{dx} = 3x^2 + 2 > 0 \text{ for all } x.$ $\therefore$ No stationary pts and $f(x)$ is always increasing. $\therefore$ cubic will only have one real root α. ii) $P(-2) = (-2)^3 + 2(-2) + 4 = -8$ $P(-1) = (-1)^3 + 2(-1) + 4 = 1$ hence, $-2 < \alpha < -1$ as the function values alternate in sign. iii) $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ where $x_0 = -1$. $x_1 = -1 - \frac{1}{5}$ $= -1.2$	2 marks suggests 2 steps or 2 reasons 1. Stationary 3. Some comment about shape of cubic. Eg increasing 1 mark Well done Well done. 1 mark substitution correct 1 mark

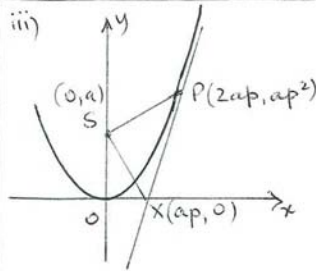
Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s) QUESTION 13	comments
b) $v^2 = 24 - 8x - 2x^2$ i) At rest $\rightarrow v=0$ ① mark $2(12 - 4x - x^2) = 0$ $x^2 + 4x - 12 = 0$ $(x+6)(x-2) = 0$ $\therefore x = -6 \text{ or } x = 2$ ① mark ii) $\frac{d}{dx}(\frac{1}{2}v^2) = \frac{d}{dx}(12 - 4x - x^2)$ $\therefore a = -4 - 2x$ ① mark iii) For max speed, $a=0$ $\therefore -4 - 2x = 0$ $x = -2$ ① mark $v^2 = 24 + 16 - 8$ $= 32$ $\therefore v = \pm \sqrt{32} = \pm 4\sqrt{2}$ ii max speed is $4\sqrt{2}$ ① mark c) $x = 20t \cos \theta$ $y = 20t \sin \theta - 5t^2 + 175$ when $t=0$, $v=20$, $\theta=30^\circ$ ① mark $\therefore x = 10\sqrt{3}t$ $y = 10t - 5t^2 + 175$ i) when $y=0$, $5t^2 - 10t - 175 = 0$ ① mark $5(t^2 - 2t - 35) = 0$ $5(t-7)(t+5) = 0$ $t = 7 \text{ or } t = -5$ But, $t \geq 0$, $\therefore t = 7$ seconds. ① mark ii) $\dot{x} = 10\sqrt{3}$ ① mark $v^2 = (\dot{x})^2 + (\dot{y})^2$ $\dot{y} = 10 - 10t$ $= (10\sqrt{3})^2 + (-40)^2$ when $t=7$, $\dot{y} = -60$ $= 300 + 3600$ ① mark $v^2 = 3900 \rightarrow v = 10\sqrt{39} \text{ m/s.}$	Well done Some students unaware of $\ddot{x} = \frac{d}{dt}(\frac{1}{2}v^2)$ Some confused $\frac{dv}{dx}$ with $\frac{dv}{dt}$ Some students said at ground level $y=175$, instead of 0 as shown in diagram & equation. Some did not obtain equations for $\ddot{x}$ and $\dot{y}$ to enable use of $v^2 = \ddot{x}^2 + \dot{y}^2$

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Suggested solution(s) QUESTION 14	comments
a) i) $\cos x - \sqrt{3} \sin x \equiv R \cos(x + \alpha)$ $= R \cos x \cos \alpha - R \sin x \sin \alpha$ $R^2 = 1 + (\sqrt{3})^2$ $= 4$ $R = 2$ ① mark $R \cos \alpha = 1$ $R \sin \alpha = \sqrt{3}$ $\therefore \tan \alpha = \sqrt{3}$ $\therefore \alpha = \frac{\pi}{3}$ (since $0 \leq \alpha < \frac{\pi}{2}$) ① mark ie. $\cos x - \sqrt{3} \sin x = 2 \cos(x + \frac{\pi}{3})$ ii) $2 \cos(x + \frac{\pi}{3}) = 1$ ① mark $\cos(x + \frac{\pi}{3}) = \frac{1}{2}$ and $0 \leq x \leq 2\pi$ $\therefore \frac{\pi}{3} \leq x + \frac{\pi}{3} \leq 2\pi + \frac{\pi}{3}$ $\therefore x + \frac{\pi}{3} = \frac{\pi}{3} \text{ or } \frac{5\pi}{3} \text{ or } \frac{7\pi}{3}$ $x = 0 \text{ or } \frac{4\pi}{3} \text{ or } 2\pi$ ① mark b) $x^2 = 4ay \rightarrow y = \frac{x^2}{4a}$, $y' = \frac{2x}{4a}$ i) at P, $x = 2ap$, $y' = \frac{4ap}{4a} = p$ ① mark Eqn of tangent at P is $y - ap^2 = p(x - 2ap)$ ① mark $y - ap^2 = px - 2ap^2$ $\therefore px - y - ap^2 = 0$ ii) $X(-, 0)$. Put $y=0$. $\therefore px = ap^2$ $x = ap$ ① mark ie X is $(ap, 0)$	Well done. Most students did not expand the domain and lost 2π solution Well done by those who attempted it.

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	QUESTION 14 continued.	comments
iii)  $m(PX) = \frac{ap^2 - 0}{2ap - ap} = p$ $m(SX) = \frac{a - 0}{0 - ap} = -\frac{1}{p}$ Since $p \cdot -\frac{1}{p} = -1$, $PX \perp SX$ (1 mark) iv) PS is diameter as $\angle PXS = 90^\circ$ (angle in semicircle) (1 mark) $\therefore$ Centre of circle C = coords of midpoint PS $= \left(\frac{2ap}{2}, \frac{ap^2 + a}{2} \right)$ (1 mark) $\therefore C = \left(ap, \frac{a(p^2 + 1)}{2} \right)$	Some students did not realise PS must be diameter. Not many attempts made at this easy question.	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	QUESTION 14 a)	comments
i) In $\triangle PXY$, $\frac{h}{PX} = \tan 20^\circ$ $\therefore PX = \frac{h}{\tan 20^\circ}$ (1 mark) ii) In $\triangle QXY$, $\frac{h}{QX} = \tan 35^\circ$ $QX = \frac{h}{\tan 35^\circ}$ (1 mark) iii) $\angle PXQ = 90^\circ$ $PX^2 + QX^2 = 40^2$ $\frac{h^2}{\tan^2 20^\circ} + \frac{h^2}{\tan^2 35^\circ} = 1600$ (1 mark) $h^2 \left(\frac{1}{\tan^2 20^\circ} + \frac{1}{\tan^2 35^\circ} \right) = 1600$ $h = \frac{\sqrt{1600}}{\sqrt{\frac{1}{\tan^2 20^\circ} + \frac{1}{\tan^2 35^\circ}}}$ $= \frac{40}{\sqrt{\frac{1}{\tan^2 20^\circ} + \frac{1}{\tan^2 35^\circ}}}$ $= \frac{40}{\sqrt{9.5882389}}$ $= 12.91785..$ $h \approx 13$ metres (1 mark)		